

firm regardless of AI exposure. For workers aged 22-25, we find a 15 log-point decline in relative employment for the most AI-exposed quintiles compared to the least exposed quintile, a large and statistically significant effect. Estimates for other age groups are much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

Fifth, the labor market adjustments are visible in employment more than in compensation. In contrast to our findings for employment, we find little difference in annual salary trends by age or exposure quintile, suggesting possible wage stickiness. If so, AI may have larger effects on employment than on wages, at least initially, or even that AI may boost wages for as many workers as it hurts.

Sixth, the above facts are largely consistent across sample constructions designed to address various alternative explanations for the core findings. Our results are not driven solely by computer occupations or by occupations susceptible to remote work and outsourcing. We also find that the AI exposure taxonomy did not meaningfully predict employment outcomes for young workers further back in time, before the widespread use of LLMs, including during the unemployment spike driven by the COVID-19 pandemic. The patterns we observe in the data appear most acutely starting in late 2022 and early 2023, around the time of rapid proliferation of generative AI tools.³ They hold for both occupations with a high share of college graduates and ones with a low college share, suggesting deteriorating education outcomes during COVID-19 do not drive our results. For non-college workers, we find evidence that experience may serve as less of a buffer to labor market disruption, as low college share occupations exhibit divergent employment outcomes by AI exposure up to age 40.

While we explore a variety of alternative explanations, we caution that the facts we document may in part be influenced by factors other than generative AI. Taken as a whole, our results are consistent with the hypothesis that generative AI has begun to affect entry-level employment. We intend to continue to track the data on an ongoing basis to assess whether these trends change in the future.

³In particular, OpenAI introduced ChatGPT in November 2022. It was reported to have over 100 million active users by January 2023 and 1.7 billion website visitors in October 2023 (DeVon, 2023).

US and UK. In contrast, the most recent version of Humlum and Vestergaard (2025) found minimal effects on entry-level earnings or hours worked in Denmark.¹¹

3 Data Description

3.1 Payroll Data

This study uses data from ADP, the largest payroll processing firm in America. The company provides payroll services for firms employing over 25 million workers in the US. We use this to track employment changes for workers in occupations measured as more or less exposed to artificial intelligence.

We make several restrictions for our main analysis sample. We include only workers with positive earnings, exclude part-time employees, and subset to people under age 70.¹²

The set of firms using payroll services changes over time as companies join or leave ADP’s platform. We maintain a consistent set of firms by keeping only companies that have employee earnings records for each month from January 2021 through September 2025.

ADP observes job titles for about 70% of workers in its system. We exclude workers without a recorded job title. There are over 7,000 standardized job titles, including “Search engineer optimization specialist,” “Enterprise content management manager,” and “Plant documentation control specialist.” The company’s internal research team maps each of these job titles to a 2010 Standard Occupational Classification (SOC) code, additionally using information such as the job description, industry, location, and other relevant data. We use these estimated SOC codes to merge our data to occupational AI exposure measures.

After these restrictions we have records on 3.5 and 5 million workers each month for our main analysis sample, though we consider robustness to alternative analyses such as allowing for firms to enter and leave the sample.

While the ADP data include millions of workers in each month, the distribution of firms using

¹¹An important direction for future work is to reconcile differences among these recent studies, assessing the extent they arise from differences in labor market institutions, measurement of AI adoption, statistical methodology, or other factors.

¹²While we observe the year of birth for each worker, for privacy reasons we do not observe the exact date of birth. We impute month of birth from the distribution of birth months in the United States using data from the Center for Disease Control and Prevention.

51 and 5415). Results are quite similar, consistent with the above case studies showing employment changes across various occupations. Results with firm-time fixed effects in Figure 4 further show that our findings are robust to firm- or industry-level shocks that impact general hiring trends. These results indicate that our findings are not specific to technology roles.

Remote Work Figures A14 and A15 show results for occupations amenable to remote work (telework) and those that are not, according to Dingel and Neiman (2020).²³ We find that, for young workers, more exposed occupations have slower employment growth, both in teleworkable occupations and in non-teleworkable occupations. The results for non-teleworkable occupations in particular suggest that our findings are not driven by outsourcing or work-from-home disruptions, at least solely.²⁴

Longer Sample Figure A16 shows results when extending the balanced sample of firms to 2018. This reduces the sample size and makes the data somewhat noisier. Nonetheless, the trends remain largely ordered by exposure in the post-GPT era, whereas this is not the case before 2022. A concern is that for the Eloundou et al. (2024) measures the most exposed quintile had slower employment growth starting around 2020. This is not the case for the Anthropic exposure measures, shown in Figures A17, A18, and A19. For these measures the most exposed groups have comparable employment growth throughout the period before generative AI, with divergent trends afterwards.

Changes in Education Another possibility is that our results stem from worsening education outcomes during the COVID-19 Pandemic, as more educated workers have greater average AI exposure (Kuhfeld and Lewis, 2025).²⁵ In Figure A20 we show trends for occupations in which greater than 70% of workers have a college degree according to the 2017 American Community Survey (ACS).²⁶ In Figure A21 we show trends for occupations in which fewer than 30% of workers

²³Whether an occupation is amenable to remote work is positively correlated with AI exposure. Only two teleworkable occupations fall in the lowest quintile of estimated AI exposure according to the GPT-4 β measure. Likewise few non-teleworkable occupations fall in the highest quintile of AI exposure. For this reason in Figure A14 we pool together the two lowest AI exposure quintiles into one group. In Figure A15 we pool together the two highest AI exposure quintiles.

²⁴Non-teleworkable occupations with high AI exposure include bank tellers, travel agents, and tax preparers.

²⁵Chandar (2025a) finds that declines in average skill levels for college graduates explain a sizable share of the slowdown in the growth of the college wage gap in recent decades.

²⁶Not a single occupation in the first quintile of GPT-4 β based exposure measure has a college share above 35%, so that quintile is excluded from the results in Figure A20.

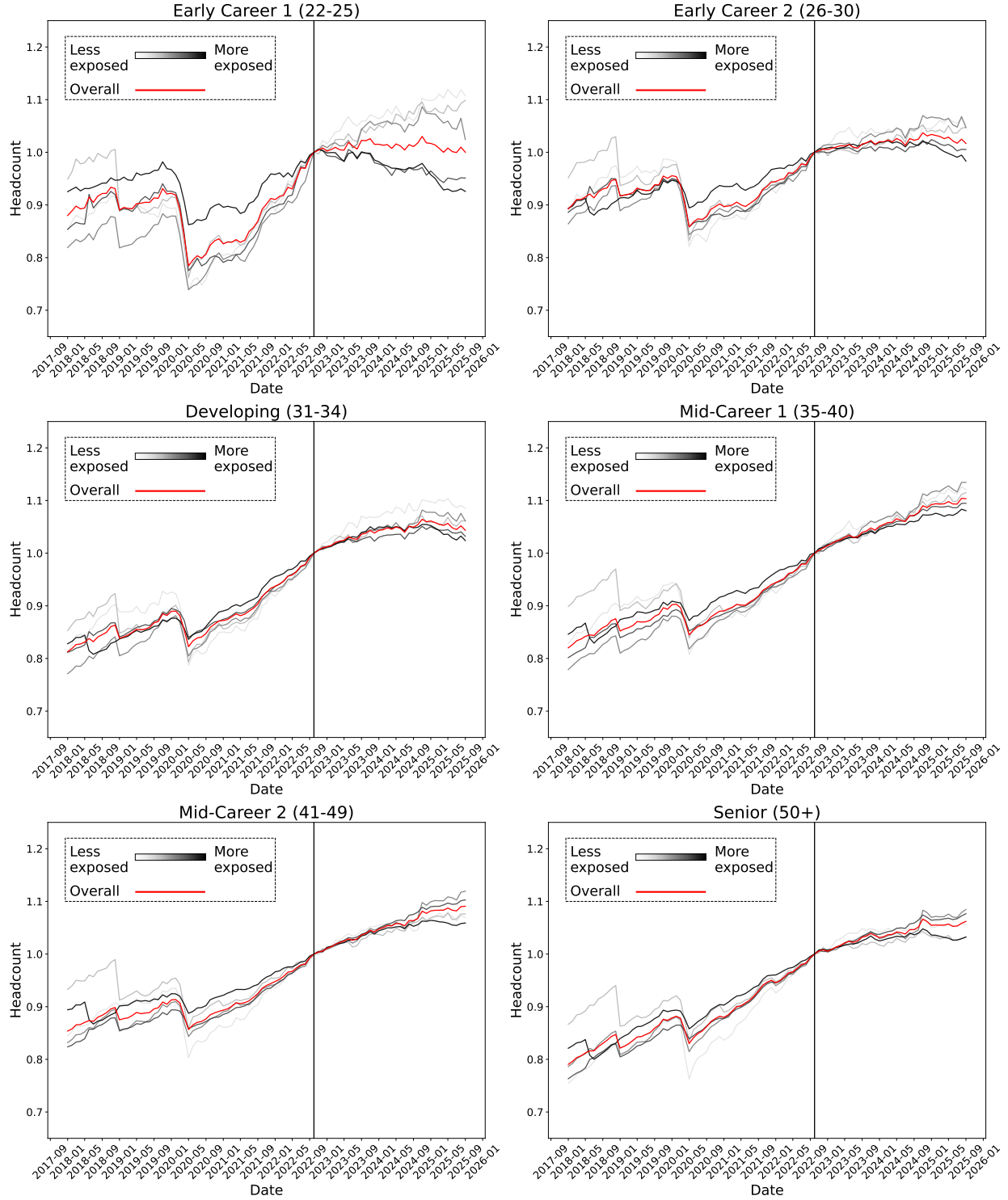


Figure A16: Employment changes by age and exposure quintile using measures from [Eloundou et al. \(2024\)](#). Data is from 2018 to 2025.

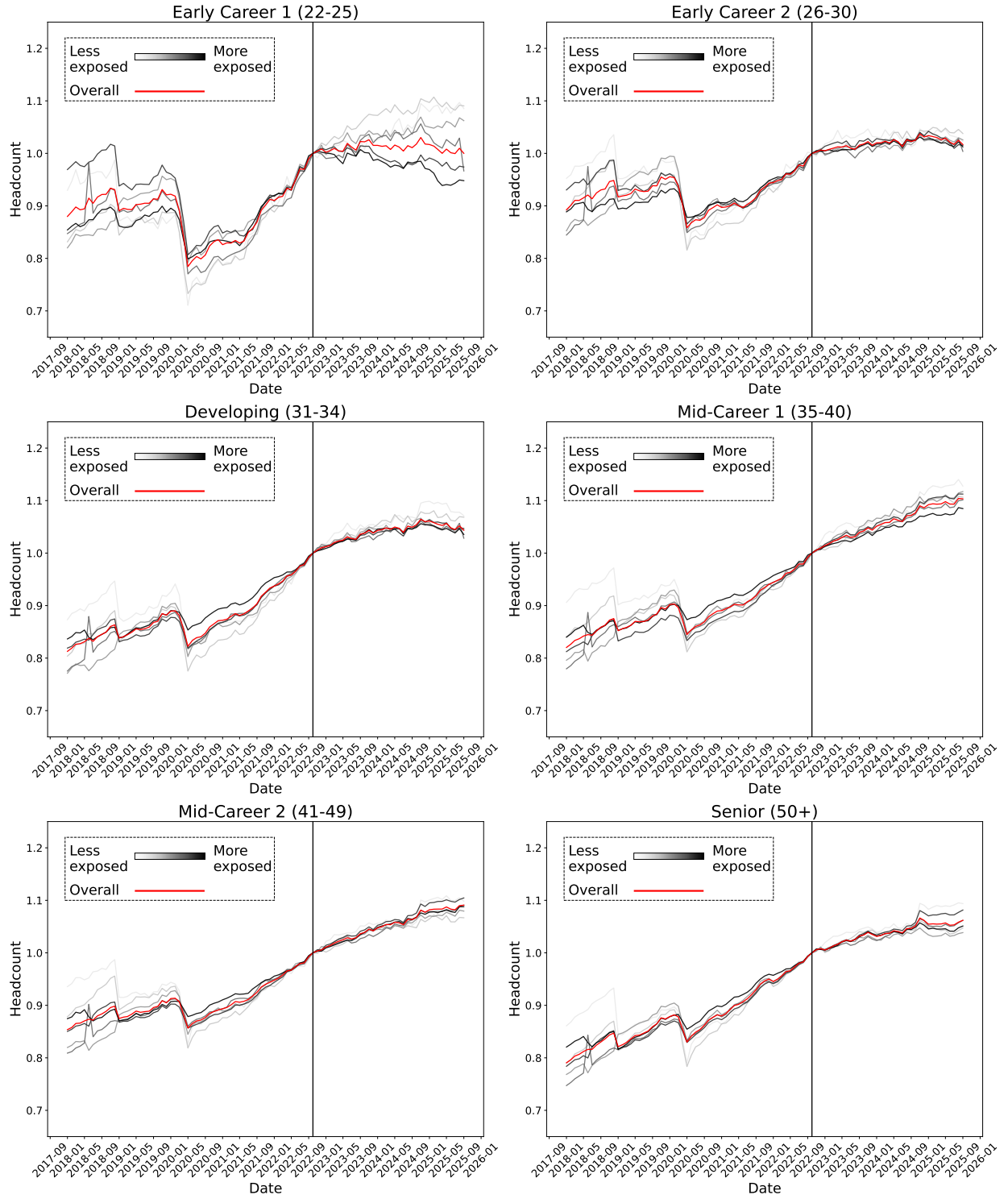


Figure A17: Employment changes by age and exposure quintile using Claude usage data from [Handa et al. \(2025\)](#). Occupations whose associated tasks all have fewer than the minimum number of queries to appear in the usage data are treated as a separate category, coded as 0. Data is from 2018 to 2025.

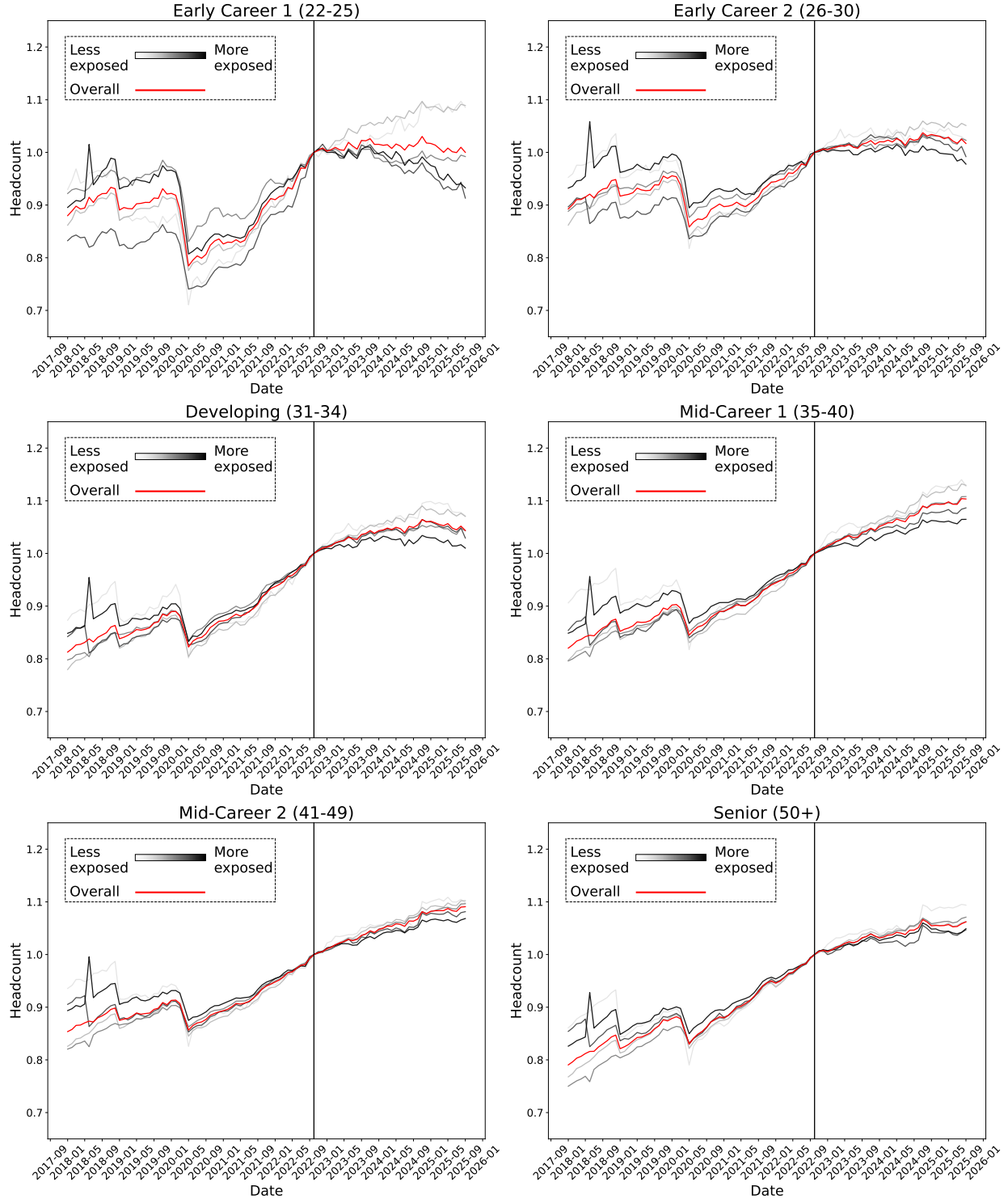


Figure A18: Employment changes by age and automation level using Claude usage data from [Handa et al. \(2025\)](#). Data is from 2018 to 2025.

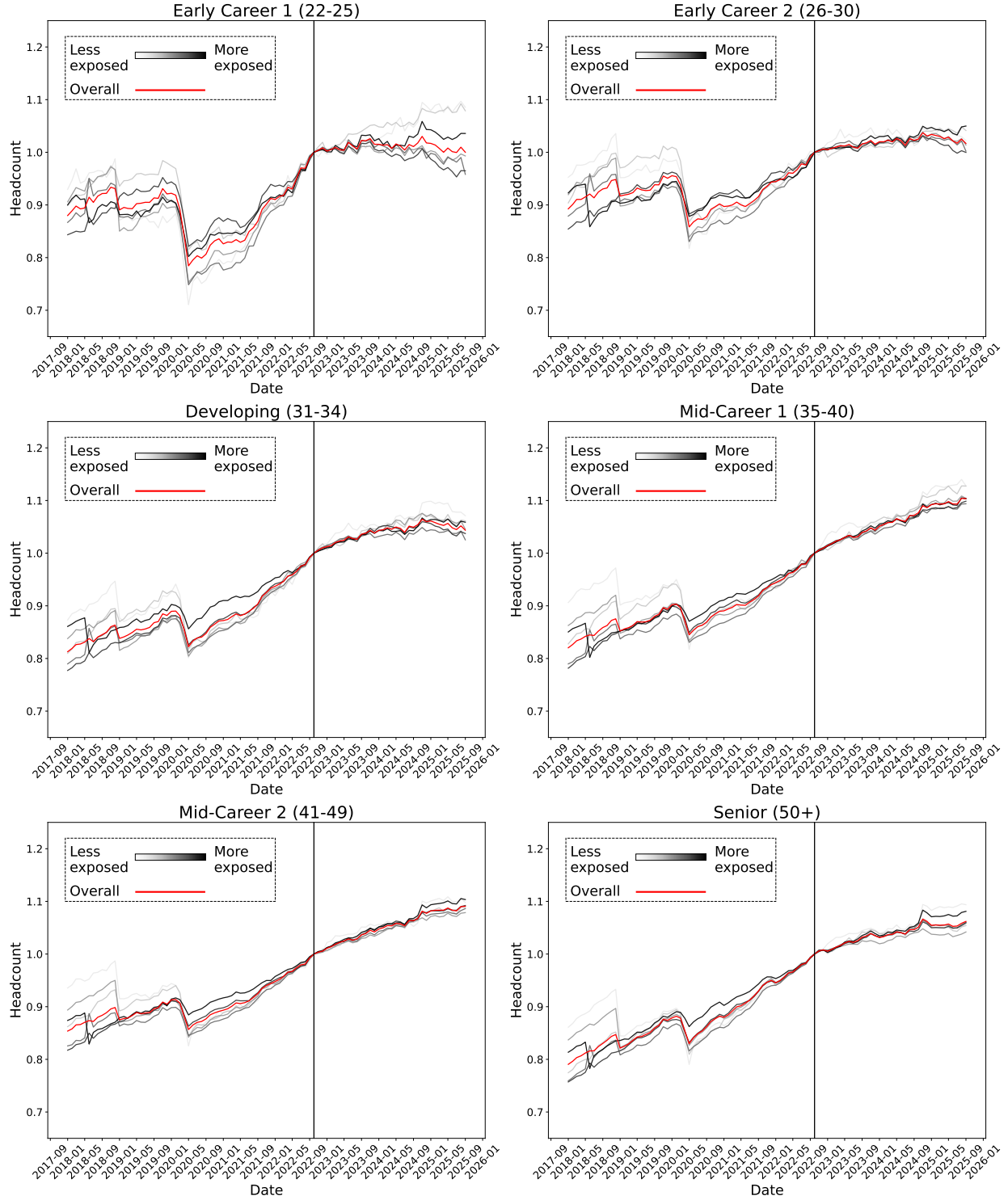


Figure A19: Employment changes by age and augmentation quintile using Claude usage data from Handa et al. (2025). Data is from 2018 to 2025.